

Course Syllabus:

Unit No.	Topic Name	Sub Topic	No. of Hours	CO Code
Unit 1	Introduction to Artificial Intelligence	Meaning, Characteristics and Objectives of AI; History and Evolution of AI - Dartmouth Conference, Major Milestones, AI Winters and Resurgence; The Turing Test and the Question of Machine Intelligence; Human Intelligence vs. Machine Intelligence; Symbolic (Rule-Based) AI vs. Data-Driven (Learning-Based) AI; Hierarchy and Taxonomy of AI, ML, Deep Learning and Generative AI; Types of AI - Narrow AI, General AI and Super AI; Weak AI vs. Strong AI; Tasks Performed by AI - Mundane Tasks, Formal Tasks and Expert Tasks; AI Capabilities and Limitations; AI Myths vs. Realities; AI in Everyday Life; AI vs. Automation; Interdisciplinary Scope of AI - Applications Across Healthcare, Business, Law and Marketing	10	CO1
Unit 2	Intelligent Systems and Problem Solving	Intelligent Systems - Basic Concept; Intelligent Agents and Rational Agents; Agent, Environment and the Agent Function; PEAS Description - Performance Measure, Environment, Actuators, Sensors; Perception and Action Cycle; Types of Environments - Fully vs. Partially Observable, Deterministic vs. Stochastic, Episodic vs. Sequential, Static vs. Dynamic, Discrete vs. Continuous, Single-Agent vs. Multi-Agent; Types of Agents - Simple Reflex, Model-Based Reflex, Goal-Based, Utility-Based and Learning Agents; Goal-Oriented Behaviour; AI-Based Problem Solving - Problem Formulation, State, Goal and Actions; Knowledge Representation and Rules - Basic Concept; Reasoning and Inference - Basic Concept	9	CO2
Unit 3	Introduction to Search in AI	Problem Formulation - Initial State, Goal State and Path; Production Systems - Rules and Control Strategy; State Space - Representing a Problem as a Set of States; Search Trees and Search Space; Types of Search - Uninformed (Blind) Search vs. Informed (Heuristic) Search; Basic Uninformed Search Algorithms - Breadth-First Search and Depth-First Search; Concept of a Heuristic; Basic Informed Search Algorithms - Greedy Best-First Search and A* Search - Basic Idea; Search in Game Playing - Basic Idea; Real-World Applications of Search Algorithms - Route Planning, Scheduling, Recommendation Ranking	9	CO3

Unit No.	Topic Name	Sub Topic	No. of Hours	CO Code
Unit 4	AI Technologies	Machine Learning - Introduction; Deep Learning - Introduction; Natural Language Processing; Computer Vision; Speech and Voice AI; Expert Systems; Recommendation Systems; Robotics and Autonomous Systems; Conversational AI; Introduction to Generative AI; Relationship Among Major AI Technologies	9	CO4
Unit 5	AI Ethics, Governance and Future of Work	Introduction to AI Ethics — Need and Importance; Algorithmic Bias and Fairness; Privacy and Data Protection Concerns with AI; Deepfakes and Synthetic Media; Responsible AI Principles and Overview of Regulations (e.g., EU AI Act, India's DPDP Act); Future of Work - Human-AI Collaboration; Augmentation vs. Displacement; Critical Thinking and Human Oversight in an AI-Driven World	8	CO5
Total			45 hours	

Detailed Lecture Plan:

Week No.	Session ID	Session Type	Topic Name	Sub-topic	Topic Description	Book ID	Reading Resources	Is in Class Practical	CO Code
1	L1	Lecture	Meaning, Characteristics and Objectives of AI	Meaning, Characteristics and Objectives of AI	Defines AI, its core characteristics, and the objectives it aims to achieve.	T1	RR1	No	CO 1
1	L2	Lecture	History and Evolution of AI	History and Evolution of AI	Dartmouth Conference, major milestones, and the cycles of AI Winters and Resurgence.	T1, R1	RR2	No	CO 1



Week No.	Session ID	Session Type	Topic Name	Sub-topic	Topic Description	Book ID	Reading Resources	Is in Class Practical	CO Code
1	T1	Tutorial	History and Evolution of AI	Timeline Construction of AI History	Constructing a visual timeline of AI's key milestones, from the Dartmouth Conference to the present day.	T1	RR2	No	CO 1
2	L3	Lecture	The Turing Test and the Question of Machine Intelligence ; Human Intelligence vs. Machine Intelligence	The Turing Test and Machine Intelligence ; Human vs. Machine Intelligence	Explores the Turing Test and whether machines can think, and compares human and machine cognition.	T1, R1	RR1, RR2	No	CO 1
2	L4	Lecture	Symbolic (Rule-Based) AI vs. Data-Driven (Learning-Based) AI; Hierarchy and Taxonomy of AI, ML, Deep Learning and Generative AI	Symbolic vs. Data-Driven AI; Hierarchy and Taxonomy of AI	Contrasts rule-based (symbolic) AI with data-driven (learning-based) AI, and explains how AI, ML, Deep Learning and Generative AI are nested and related.	T1	RR1	No	CO 1
2	T2	Tutorial	Symbolic (Rule-Based) AI vs. Data-Driven (Learning-Based) AI	Symbolic vs. Data-Driven Case Comparison	Comparing symbolic (rule-based) and data-driven (learning-based) approaches using simple real-world case examples.	T1	RR1	No	CO 1

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3	L5	Lecture	Types of AI; Weak AI vs. Strong AI	Types of AI: Narrow, General, Super; Weak AI vs. Strong AI	Classifies AI by scope (Narrow/General/Super) and by sophistication (Weak/Strong).	T1, R1	RR2	No	CO 1
3	L6	Lecture	Tasks Performed by AI; AI Capabilities and Limitations	Tasks Performed by AI; AI Capabilities and Limitations	Covers Mundane, Formal and Expert Tasks performed by AI, along with what AI can and cannot currently do.	T1	RR1	No	CO 1
3	L7	Lecture	AI Myths vs. Realities; AI in Everyday Life; AI vs. Automation ; Interdisciplinary Scope of AI	AI Myths vs. Realities; AI in Everyday Life; AI vs. Automation ; Interdisciplinary Scope	Separates common misconceptions from facts, situates AI in daily life, distinguishes it from automation, and surveys its use across Healthcare, Business, Law and Marketing.	T1, R1	RR6	No	CO 1
4	T3	Tutorial	Hierarchy and Taxonomy of AI, ML, Deep Learning and Generative AI; Types of AI; AI in Everyday Life	Classifying Real-World AI Systems ("AI Around Me")	Identifying AI applications from everyday life (voice assistants, recommendation engines, self-driving cars) and classifying them by type and taxonomy.	T1	RR1, RR6	No	CO 1
4	L8	Lecture	Intelligent Systems - Basic	Intelligent Systems; Intelligent	Introduces intelligent systems and	T1	RR3	No	CO 2

Week No.	Session ID	Session Type	Topic Name	Sub-topic	Topic Description	Book ID	Reading Resources	Is in Class Practical	CO Code
			Concept; Intelligent Agents and Rational Agents	Agents and Rational Agents	the concept of a rational agent that acts to achieve the best outcome.				
4	L9	Lecture	Agent, Environment and the Agent Function; PEAS Description	Agent, Environment and the Agent Function; PEAS Description	Explains the agent-environment relationship and the PEAS framework (Performance measure, Environment, Actuators, Sensors).	T1	RR3	No	CO 2
5	T4	Tutorial	PEAS Description	Designing PEAS Descriptions	Designing PEAS descriptions for real-world systems such as a navigation app, chatbot or recommendation system.	T1	RR3	No	CO 2
5	L10	Lecture	Perception and Action Cycle; Types of Environments	Perception and Action Cycle; Types of Environments	Covers the perceive-act cycle and classifies environments (observable, deterministic, episodic, static, discrete, single/multi-agent).	T1	RR3	No	CO 2
5	L11	Lecture	Types of Agents	Types of Agents	Covers Simple Reflex, Model-Based Reflex, Goal-Based, Utility-Based and Learning Agents.	T1, R1	RR3	No	CO 2

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6	T5	Tutorial	Types of Agents	Classifying Agent Types for Given Scenarios	Matching real-world scenarios to the correct agent type (reflex, goal-based, utility-based, learning).	T1	RR3	No	CO 2
6	L12	Lecture	Goal-Oriented Behaviour; AI-Based Problem Solving	Goal-Oriented Behaviour; AI-Based Problem Solving	Introduces goal-directed behaviour and frames problem solving in terms of problem formulation, state, goal and actions.	T1	RR3	No	CO 2
6	L13	Lecture	Knowledge Representation and Rules - Basic Concept; Reasoning and Inference - Basic Concept	Knowledge Representation and Rules; Reasoning and Inference — Basic Concepts	Introduces how knowledge and rules are represented for intelligent systems, and how systems draw inferences from that knowledge.	T1	RR1	No	CO 2
7	T6	Tutorial	AI-Based Problem Solving	Formulating Problems in State-Space Terms	Practising problem formulation on paper by identifying initial state, goal state and possible actions for simple problems.	T1	RR1	No	CO 2
7	L14	Lecture	Problem Formulation ; Production Systems	Problem Formulation ; Production Systems	Covers initial state, goal state and path, along with production systems (rules	T1	RR5	No	CO 3



We ek No.	Sessi on ID	Sessi on Type	Topic Name	Sub-topic	Topic Description	Bo ok ID	Readi ng Resou rces	Is in Class Practi cal	CO Co de
					and control strategy).				
7	L15	Lectu re	State Space; Search Trees and Search Space	State Space; Search Trees and Search Space	Explains how a problem is represented as a state space, and introduces search trees.	T1	RR5	No	CO 3
8	T7	Tutor ial	State Space	Representi ng a Problem as a State Space	Representing a sample problem (e.g., a maze or sliding puzzle) as a state space on paper.	T1	RR5	No	CO 3
8	L16	Lectu re	Types of Search; Basic Uninformed Search Algorithms	Types of Search: Uninformed vs. Informed; BFS and DFS	Distinguishes uninformed (blind) from informed (heuristic) search, and covers Breadth-First and Depth-First Search with their trade-offs.	T1	RR4	No	CO 3
8	T8	Tutor ial	Basic Uninformed Search Algorithms	Tracing BFS/DFS Traversal	Manually tracing Breadth-First and Depth-First Search traversal on a sample graph.	T1	RR4	No	CO 3
9	L17	Lectu re	Concept of a Heuristic; Basic Informed Search Algorithms	Concept of a Heuristic; Greedy Best-First Search and A*	Introduces heuristics as educated guesses that guide search, and covers the basic idea of Greedy Best-First and A* Search.	T1	RR4	No	CO 3

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9	T9	Tutorial	Basic Informed Search Algorithms	Tracing Greedy/A* Search	Manually tracing Greedy Best-First and A* Search on a sample graph using given heuristic values.	T1	RR4	No	CO 3
9	L18	Lecture	Search in Game Playing - Basic Idea	Search in Game Playing — Basic Idea	Introduces how search is applied in simple game-playing scenarios such as Tic-Tac-Toe or Chess.	T1	RR5	No	CO 3
10	L19	Lecture	Real-World Applications of Search Algorithms	Real-World Applications of Search Algorithms	Covers applications such as route planning, scheduling and recommendation ranking.	T1	RR4	No	CO 3
10	L20	Lecture	Machine Learning - Introduction ; Deep Learning - Introduction	Machine Learning — Introduction ; Deep Learning — Introduction	Introduces the concept and workflow of machine learning, and deep learning as a neural-network-based subset of it.	T1, R1	RR7	No	CO 4
10	T10	Tutorial	Machine Learning - Introduction ; Deep Learning - Introduction	Comparing ML vs. DL	Comparing machine learning and deep learning approaches using simple illustrative examples.	T1, R1	RR7	No	CO 4
11	L21	Lecture	Natural Language Processing;	Natural Language Processing;	Introduces how AI systems process human	T1, R1	RR2, RR7	No	CO 4

Week No.	Session ID	Session Type	Topic Name	Sub-topic	Topic Description	Book ID	Reading Resources	Is in Class Practical	CO Code
			Computer Vision	Computer Vision	language and interpret visual information.				
11	T11	Tutorial	Natural Language Processing; Computer Vision	Identifying NLP/CV Use Cases	Identifying NLP and computer vision use cases in everyday applications (translation apps, photo tagging, etc.).	T1, R1	RR7	No	CO 4
11	L22	Lecture	Speech and Voice AI; Expert Systems	Speech and Voice AI; Expert Systems	Covers speech recognition and synthesis, and rule-based expert systems.	T1, R1	RR2	No	CO 4
12	L23	Lecture	Recommendation Systems; Robotics and Autonomous Systems	Recommendation Systems; Robotics and Autonomous Systems	Covers how recommendation engines work, and introduces robotics and autonomous systems.	T1, R1	RR7	No	CO 4
12	L24	Lecture	Conversational AI; Introduction to Generative AI	Conversational AI; Introduction to Generative AI	Covers chatbots and conversational agents, and introduces generative AI at a basic level.	T1, R1	RR7	No	CO 4
12	L25	Lecture	Relationship Among Major AI Technologies	Relationship Among Major AI Technologies	Synthesises how ML, DL, NLP, CV, robotics and GenAI relate to and build on one another.	T1, R1	RR7	No	CO 4
13	T12	Tutorial	Relationship Among Major AI	Concept Map of AI Technologies	Constructing a concept map linking all the AI technologies	T1, R1	RR7	No	CO 4

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			Technologies		covered in the unit and their relationships.				
13	L26	Lecture	Introduction to AI Ethics — Need and Importance; Algorithmic Bias and Fairness	Introduction to AI Ethics; Algorithmic Bias and Fairness	Introduces why ethics matters in AI, and covers how bias enters AI systems and approaches to fairness.	R2	RR8	No	CO 5
13	T13	Tutorial	Algorithmic Bias and Fairness	Bias Identification Exercise	Identifying algorithmic bias using real-world case examples from hiring, healthcare or credit scoring.	R2	RR8	No	CO 5
14	L27	Lecture	Privacy and Data Protection Concerns with AI; Deepfakes and Synthetic Media	Privacy and Data Protection; Deepfakes and Synthetic Media	Covers privacy risks and data protection in AI systems, and deepfakes and synthetic media with the risks they pose.	R2	RR9, RR10	No	CO 5
14	T14	Tutorial	Deepfakes and Synthetic Media	Deepfake Awareness Discussion	Discussing techniques for recognising deepfakes and synthetic media, and their societal risks.	R2	RR9	No	CO 5
14	L28	Lecture	Responsible AI Principles and Overview of Regulations	Responsible AI Principles and Overview of Regulations	Covers responsible AI principles and regulatory frameworks such as the EU AI Act and India's DPDP Act.	R2	RR9, RR10	No	CO 5

Week No.	Session ID	Session Type	Topic Name	Sub-topic	Topic Description	Book ID	Reading Resources	Is in Class Practical	CO Code
15	L29	Lecture	Future of Work — Human-AI Collaboration; Augmentation vs. Displacement	Future of Work: Human-AI Collaboration; Augmentation vs. Displacement	Explores how AI is reshaping work through collaboration, and debates augmentation versus displacement of jobs.	R2	RR6	No	CO 5
15	T15	Tutorial	Future of Work — Human-AI Collaboration; Augmentation vs. Displacement	Debate: Augmentation vs. Displacement	Structured debate on whether AI will augment or displace jobs in specific industries.	R2	RR6	No	CO 5
15	L30	Lecture	Critical Thinking and Human Oversight in an AI-Driven World	Critical Thinking and Human Oversight in an AI-Driven World	Emphasises the need for human oversight and critical thinking when working with AI-driven systems.	R2	RR8	No	CO 5